Recursion Doubt
Kartik 10:00 AM

Print numbers from 1 to N
Example:
Input: N = 5
Output: 1 2 3 4 5
Q2. Print numbers from N to 1
Example:
Input: N = 5
Output: 5 4 3 2 1
Q3. Sum of first N natural numbers
Example:
Input: N = 5
Output: 15
Q4. Factorial of N
Example:
Input: N = 5
Output: 120
Q5. Sum of digits
Example:
Input: N = 123
Output: 6
Q6. Count digits
Example:
Input: N = 9876
Output: 4
Q7. Product of digits
Example:
Input: N = 234
Output: 24
Q8. Reverse a number
Example:
Input: N = 123
Output: 321
Q9. Check palindrome number

1 2 1

Q10. Power a^b
Example:
Input: a = 2, b = 4
Output: 16
Q11. Maximum element in array
Example:
Input: [1, 5, 3]
Output: 5
Q12. Minimum element in array
Example:
Input: [1, 5, 3]
Output: 1
Q13. Sum of array elements
Example:
Input: [1, 2, 3]
Output: 6
Q14. Check array sorted
Example:
Input: [1, 2, 3]
Output: True
Q15. Count element occurrences
Example:
Input: [1, 2, 2, 3], key = 2
Output: 2
Q16. First occurrence in array
Example:
Input: [1, 2, 3, 2], key = 2
Output: index 1
Q17. Last occurrence in array
Example:
Input: [1, 2, 3, 2], key = 2
Output: index 3
Q18. Print array elements
Example:

Q19. Count even numbers
Example:
Input: [1, 2, 4]
Output: 2
Q20. Count vowels in string
Example:
Input: "coding"
Output: 2
Q21. Check prime
Example:
Input: N = 7
Output: True
Q22. GCD of two numbers
Example:
Input: a = 12, b = 18
Output: 6
Q23. LCM of two numbers
Example:
Input: a = 4, b = 6
Output: 12
Q24. Sum of even digits
Example:
Input: 1234
Output: 6
Q25. Remove zeros from number
Example:
Input: 1020
Output: 12
Q26. Count zeros in number
Example:
Input: 10020
Output: 3

```
public static int count1(int [][arr,int i,int c) {
    if(i==arr.length) {
        return c;
    }
    if(arr[i][0]==0) {
        c++;
    }
    return count2(arr, i+1, c);
}
```

```
public static int Countvowels1(String s) {
    if (s.length()==0) {
        return 0;
    }
    int c = 0;
    if (isvowel(s.charAt(0))) {
        c++;
    }
    return c + Countvowels1(s.substring(1));
}
```

Q28. Decimal to binary
Example:
Input: 5
Output: 101
Q29. String palindrome
Example:
Input: "madam"
Output: True
Q30. Length of string
Example:
Input: "hello"
Output: 5
Q31. Replace character
Example:
Input: "cat", a → o
Output: "cot"
Q32. Remove duplicates
Example:
Input: "aabb"
Output: "ab"
Q33. Count words
Example:
Input: "I love code"
Output: 3
Q34. Print all indexes
Example:
Input: [1, 2, 2], key = 2
Output: 1 2

```
public static void duplicates(String ques, String ans) {
    if (ques.length()==0 || ques.charAt(0)==null) {
        System.out.println(ans);
        return;
    }
    if (ans.length()==0 || ques.charAt(0) != ans.charAt(ans.length()-1)) {
        ans = ans + ques.charAt(0);
    }
    duplicates(ques.substring(1), ans);
}
```

Q35. Check strictly increasing
Example:
Input: [1, 2, 3]
Output: True
Q36. Print array in reverse

Q37. Smallest digit
Example:
Input: 492
Output: 2
Q38. Largest digit
Example:
Input: 492
Output: 9
Q39. Count consonants
Example:
Input: "code"
Output: 2
Q40. Remove character x
Example:
Input: "xoxo"
Output: "oo"
Q41. Sum of alternate digits
Example:
Input: 1234
Output: 4
Q42. Check string contains only digits
Example:
Input: "1234"
Output: True